RESEARCH PROJECT & PROPOSAL APPROVAL BRIEF

Cascaded Adaptive Blended RAG: Optimizing Retrieval Efficiency via Pre-Retrieval Query Cascading

Author: Master's Student **Target Base Paper:** Sawarkar et al. (2024) **Focus:** RAG Efficiency & Adaptive Routing

1. Base Paper Selection & Motivation

Selected Base Paper: *'Blended RAG: Improving RAG (Retriever-Augmented Generation) Accuracy with Semantic Search and Hybrid Query-Based Retrievers'* (Sawarkar et al., 2024).

Why This Paper? The overall accuracy of Retrieval-Augmented Generation (RAG) systems heavily depends on retriever quality. Sawarkar et al. demonstrated that combining dense vector indexes and sparse encoder indexes with multi-field hybrid query strategies (such as *Best Fields* search across titles, abstracts, and body contexts) significantly outperforms single-retriever baselines (e.g., standard dense embeddings or BM25 keyword search) and avoids costly domain-specific model fine-tuning.

The Core Problem & Research Gap: The Static Hybrid Penalty

While Blended RAG achieves top-tier retrieval accuracy across benchmarks like NQ and HotpotQA, it suffers from a major computational bottleneck: **The Static Hybrid Penalty**. Blended RAG executes full multi-field hybrid queries (Dense Vector Search + Sparse ELSER Search + Multi-Field Expansion) for *every incoming query* regardless of difficulty. Simple factual queries that could easily be answered by a fast lexical lookup pay the exact same latency and compute cost as complex compositional queries.

2. Proposed Research Extension: Cascaded Adaptive Blended RAG

Core Objective: Develop a lightweight **Pre-Retrieval Cascaded Router** that dynamically evaluates query difficulty before search execution. Simple queries exit early via a fast BM25 pass, while difficult queries escalate to full Blended RAG (BM25 + SPLADE + Dense MiniLM).
Quantitative Target: Reduce retrieval latency and vector search calls by **40–60%** while maintaining **95%+** of full Blended RAG accuracy.

Key Architectural Components:

Pre-Retrieval Analysis — TIER 0

Extracts fast, zero-overhead features from query text alone (length, rare token count, IDF score statistics, entropy) prior to index lookup.

BM25 Fast Pass & Early Exit — TIER 1

Runs a near-zero-cost lexical search. If top-1 score $s_1 \geq \tau_{\text{high}}$ or margin $\Delta = s_1 - s_2$ is wide, returns BM25 candidates immediately, skipping vector search.

Fallback Blended Search — TIER 2

If BM25 confidence is low ($\Delta < \tau_{\text{low}}$), escalates to 3-way hybrid search: BM25 + SPLADE (Sparse Learned) + Dense MiniLM merged via Reciprocal Rank Fusion (RRF).

Metadata Query Expansion — FIELD STRATEGY

Evaluates multi-field matching (Best Fields across title, abstract, and content) to preserve Blended RAG's core innovation on structured datasets at zero added index cost.

RECENT LITERATURE (2025–2026)

DAT (Hsu et al.) & QuDAR (Kim et al.)

Focus on dynamic per-query alpha weighting (α-tuning). However, they execute both sparse and dense retrievers for 100% of queries, saving no execution latency.

OUR PROPOSED EXTENSION

Enforced Early Termination

In contrast, our proposed Cascaded Architecture enforces early termination at Tier 1, providing measurable CPU/compute savings and bypassing vector calls.

3. Methodology & System Architecture

The system pipeline processes incoming queries sequentially through early-exit logic, as illustrated in the framework mapping below:

PIPELINE STAGE	MECHANISM / MODELS USED	COMPUTE COST	TARGET QUERY TYPE
Tier 0: Pre-Retrieval	Query Length, IDF Statistics, Rare Tokens	`< 1 ms (CPU)`	All incoming user queries
Tier 1: BM25 Fast Pass	BM25 Lexical Index + Margin Check (Δ)	`~ 2–5 ms (CPU)`	Simple factual / exact-keyword match
Tier 2: Blended Hybrid	BM25 + SPLADE Sparse + Dense MiniLM + RRF	`~ 30–80 ms (CPU)`	Complex, compositional, or abstract
Field Strategy	Best Fields / Cross Fields on Title & Abstract	`Zero added cost`	Metadata-rich corpora (e.g. SciFact)

4. Experimental Setup & Evaluation Protocol

To rigorously validate the extension, experiments will be conducted on CPU-constrained laptop environments using open-source benchmarks from the BEIR suite:

- **Primary Datasets:** BEIR SciFact (5,183 scientific abstracts with title/abstract metadata) and Natural Questions (NQ).
- **Baselines to Compare:** (1) Standard BM25, (2) Dense MiniLM Vector Search, (3) Static Blended RAG (Sawarkar et al.), (4) Cascaded Adaptive Blended RAG (Proposed).

Evaluation Dimensions & Metrics:

DIMENSION	SPECIFIC METRICS	TARGET OUTCOME / BENCHMARK GOAL
Retrieval Quality	Recall@10, MRR@10, nDCG@10	`Maintain ≥ 95% of static Blended RAG accuracy`
Computational Efficiency	Latency (ms/query), % Vector Calls Avoided	`Reduce mean retrieval latency by 40–60%`
Downstream Generation	Exact Match (EM) & F1 Score (via Llama-3.2-3B)	`Ensure zero degradation in generator factual accuracy`

5. Implementation Roadmap & Milestones

PHASE	MILESTONE TASK	KEY DELIVERABLE
Phase 1 (Week 1)	Setup BEIR SciFact index; build BM25 & Dense baselines	Baseline evaluation scripts & benchmark logs
Phase 2 (Week 2)	Implement Static Blended RAG with BM25 + SPLADE + RRF	Reproduction of multi-field Blended RAG accuracy
Phase 3 (Week 3)	Develop Tier 0 query feature extractor & Tier 1 cascade router	Tuned threshold protocol ($\tau_{\text{high}}, \Delta$)
Phase 4 (Week 4)	End-to-end evaluation (Retrieval + 3B Local LLM Generation)	Final Thesis Chapter & Latency vs. Accuracy Plots

FORMAL ADVISOR REVIEW

Summary for Advisor Approval

"This research addresses an unresolved efficiency bottleneck in state-of-the-art hybrid RAG systems. It is technically grounded, reproducible on accessible hardware, and builds directly upon foundational literature (Sawarkar et al., 2024). Approval is requested to proceed with Phase 1 benchmarking."